Azure Cloud Adoption Framework (CAF) & Azure Landing Zones

a course by by Christopher Nett

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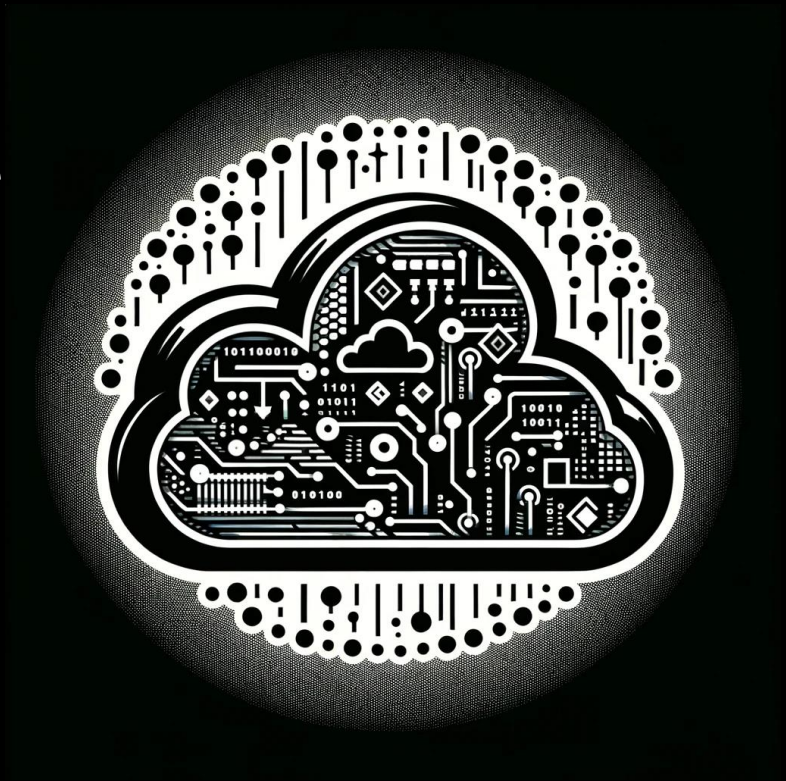
Cloud Computing Properties

- On-demand self service
- Broad network access
- Resource pooling
- Rapid elastic
- Measured service



Source: nistspecialpublication800-145.pdf

Public, Private, Multi & Hybrid Cloud

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Source: nistspecialpublication800-145.pdf

Azure Global Backbone

60+
Azure regions

130k+
miles of fiber + subsea cables

172+
Network Edge sites

500+
network partners

20k+
peering connections

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Source: [Global Network – Backbone Networking Infrastructure | Microsoft Azure]

Shared Responsibility in Azure

Responsibility	On-prem	IaaS	PaaS	SaaS
Information and Data	■	■	■	■
Devices (Mobile and PCs)	■	■	■	■
Accounts and Identities	■	■	■	■
Identity and Directory Infrastructure	■	■	◪	◪
Applications	■	■	◪	□
Network Controls	■	■	◪	□
Operating System	■	■	□	□
Physical Hosts	■	□	□	□
Physical Network	■	□	□	□
Physical Datacenter	■	□	□	□

Source: Shared responsibility in the cloud - Microsoft Azure | Microsoft Learn

Azure Subscription Types

Free

- Free credits for 30 days
- Some services are free for 12 months

Student

- Free credits for 12 months
- No credit card required

Pay As You Go

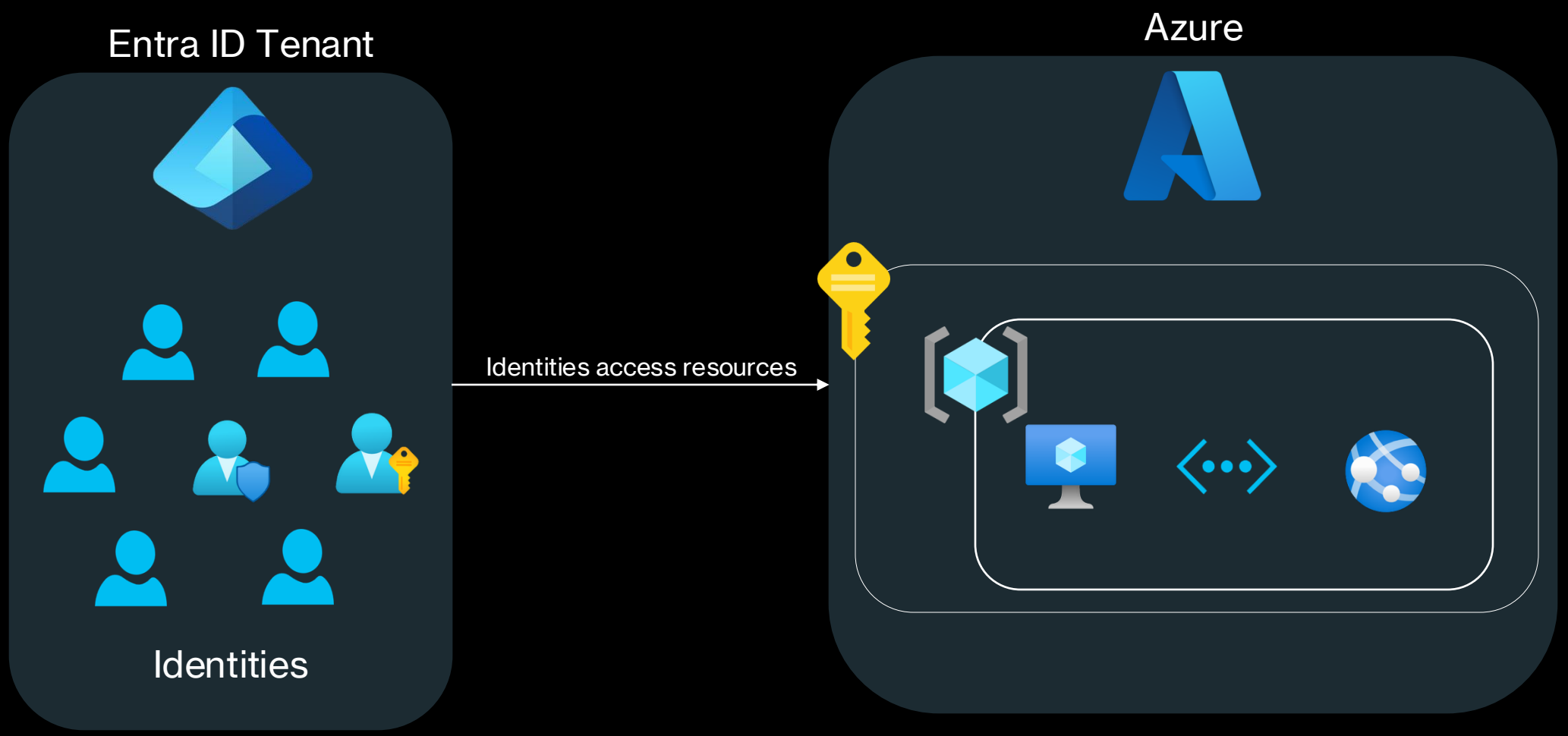
- Pay for what you use
- Credit card required

Enterprise Agreement

- One consumption agreement for all Azure services
- Various different billing models

Source: [Create Your Azure Free Account Today | Microsoft Azure]

Entra ID Tenants and Azure Subscriptions



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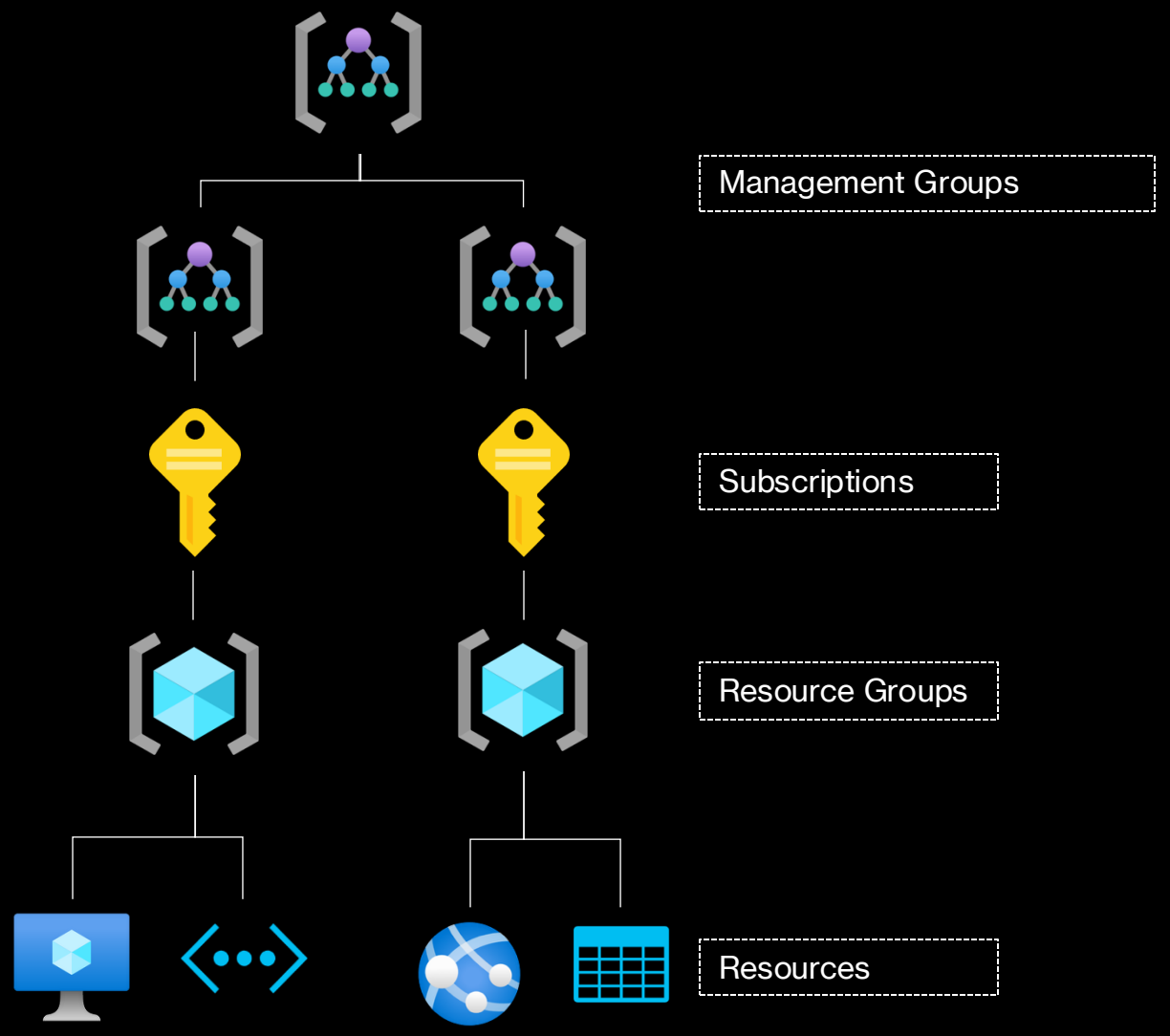
Source: [Microsoft Entra ID documentation - Microsoft Entra ID | Microsoft Learn]

Azure Resource Hierarchy



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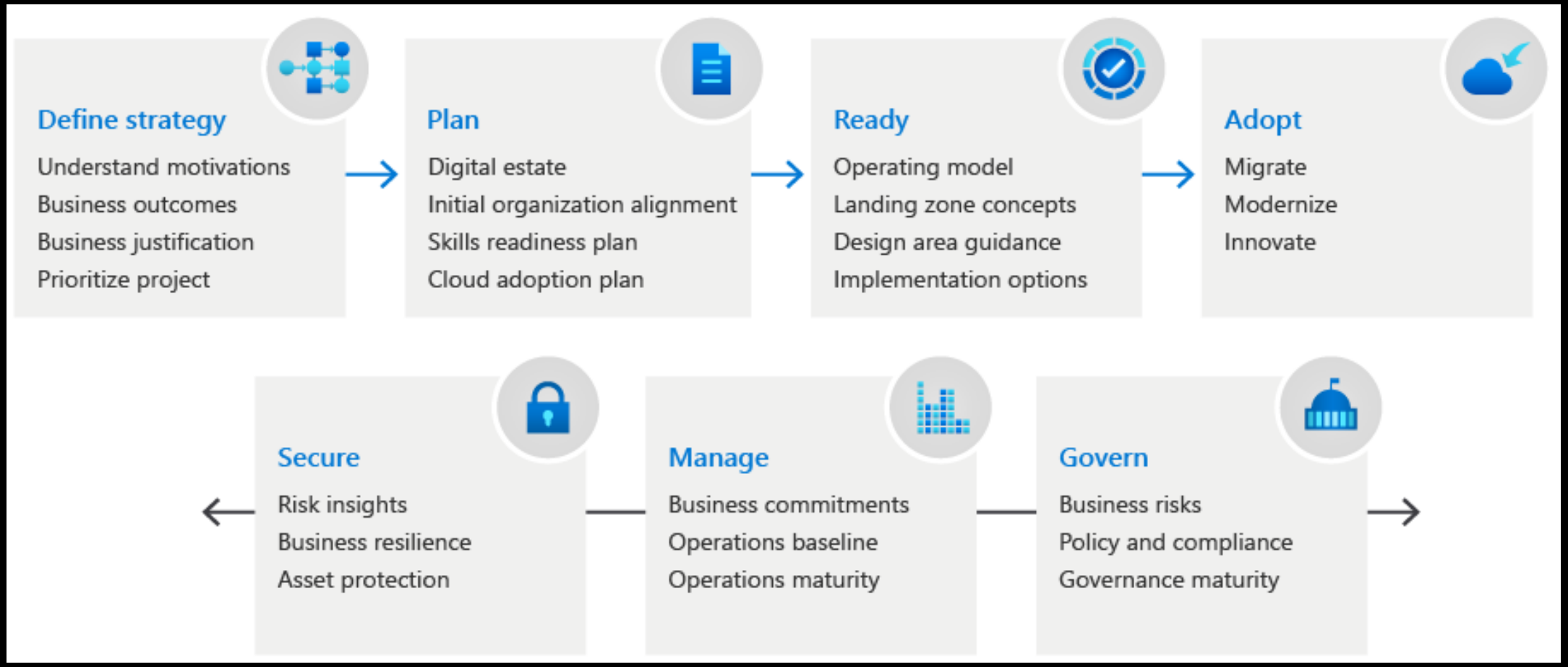
What is the Azure Cloud Adoption Framework (CAF)?

- CAF is Microsoft's end-to-end framework offering best practices, tools, and documentation to help organizations align business and technology strategies for successful cloud adoption.
- The framework is organized into key methodologies: Strategy, Plan, Ready, Migrate, Innovate, Govern, Secure, and Manage – each addressing specific phases of cloud transformation.
- CAF provides guidance tailored for business leaders, IT professionals, architects, and governance teams, fostering collaboration across departments to drive cloud success.
- CAF incorporates proven practices from Microsoft, its partners, and customers, ensuring practical and actionable insights for real-world cloud adoption scenarios.
- Beyond migration, CAF emphasizes innovation, security, governance, and ongoing management to help organizations continuously optimize and evolve their cloud environments.

Source: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

What is the Azure Cloud Adoption Framework (CAF)?



Source: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/overview

What is an Azure Landing Zone?

- An Azure landing zone is an environment that follows key design principles across certain design areas.
- These design principles accommodate all application portfolios and enable application migration, modernization, and innovation at scale.
- An Azure landing zone uses subscriptions to isolate and scale application resources and platform resources.
- Subscriptions for application resources are called application landing zones, and subscriptions for platform resources are called platform landing zones.

Source: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

What is an Azure Landing Zone?

- An Azure landing zone architecture is scalable and modular to meet various deployment needs.
- The repeatable infrastructure allows you to apply configurations and controls to every subscription consistently.
- Modules make it easy to deploy and modify specific Azure landing zone architecture components as your requirements evolve.
- The Azure landing zone conceptual architecture represents an opinionated target architecture for your Azure landing zone. You should use this conceptual architecture as a starting point and tailor the architecture to meet your needs.

Source: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

Azure Landing Zones - Design Principles

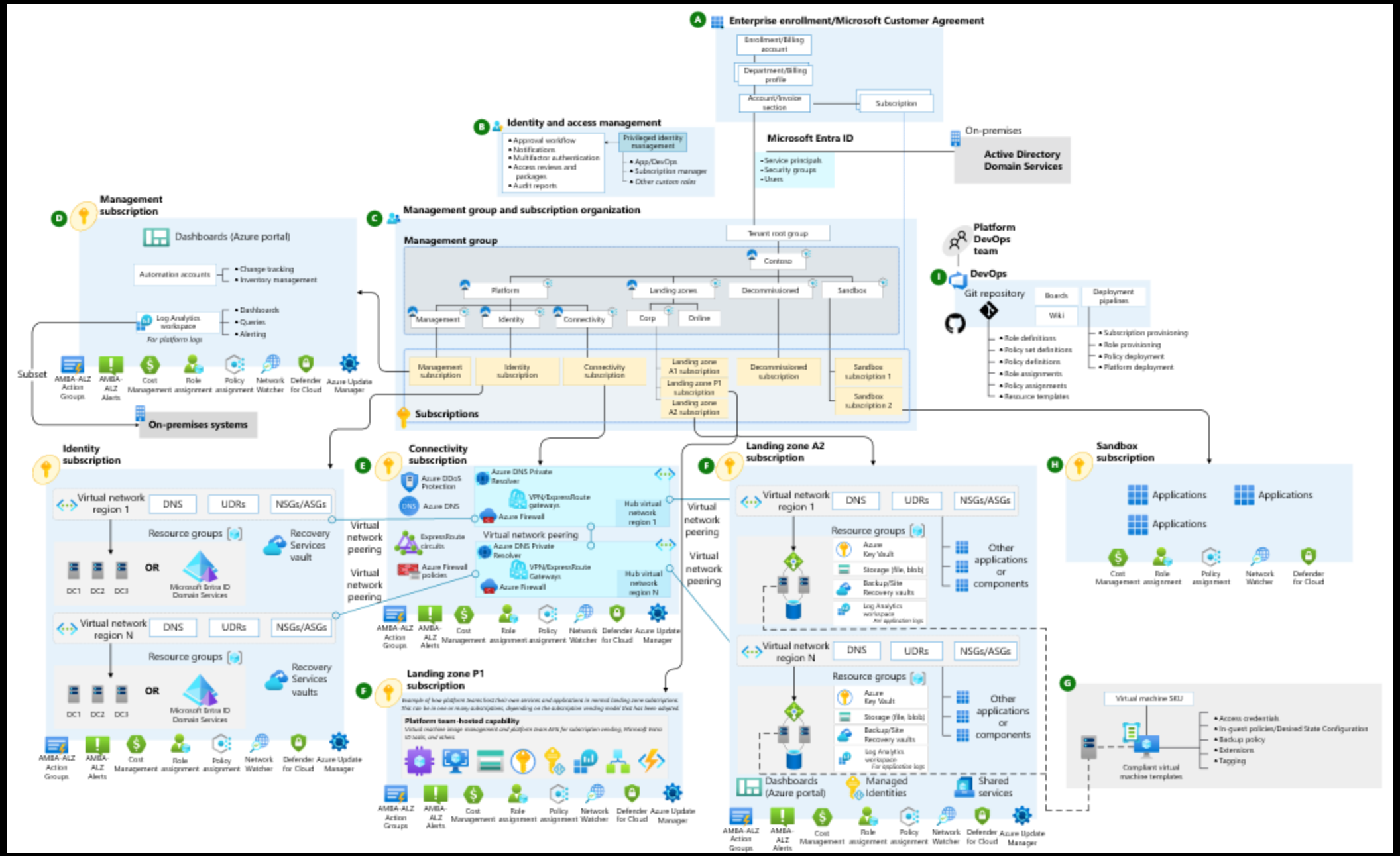
Design Area	Objective
Azure Billing and Entra tenant management	Proper tenant creation, enrollment, and billing setup are important early steps.
Identity and access management	Identity and access management is a primary security boundary in the public cloud. It's the foundation for any secure and fully compliant architecture.
Network topology and connectivity	Networking and connectivity decisions are an equally important foundational aspect of any cloud architecture.
Resource organization	As cloud adoption scales, considerations for subscription design and management group hierarchy have an impact on governance, operations management, and adoption patterns.

Source: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/design-areas#environment-design-areas

Azure Landing Zones - Design Areas

Design Area	Objective
Security	Implement controls and processes to protect your cloud environments.
Management	For stable, ongoing operations in the cloud, a management baseline is required to provide visibility, operations compliance, and protect and recover capabilities.
Governance	Automate auditing and enforcement of governance policies.
Platform automation & DevOps	Align the best tools and templates to deploy your landing zones and supporting resources.

Source: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/design-areas#environment-design-areas



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Source: https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/

Platform landing zones vs. application landing zones

An Azure landing zone consists of platform landing zones and application landing zones.

- Platform landing zone: A platform landing zone is a subscription that provides shared services (identity, connectivity, management) to applications in application landing zones. Consolidating these shared services often improves operational efficiency.
- Application landing zone: An application landing zone is a subscription for hosting an application. You pre-provision application landing zones through code and use management groups to assign policy controls to them.

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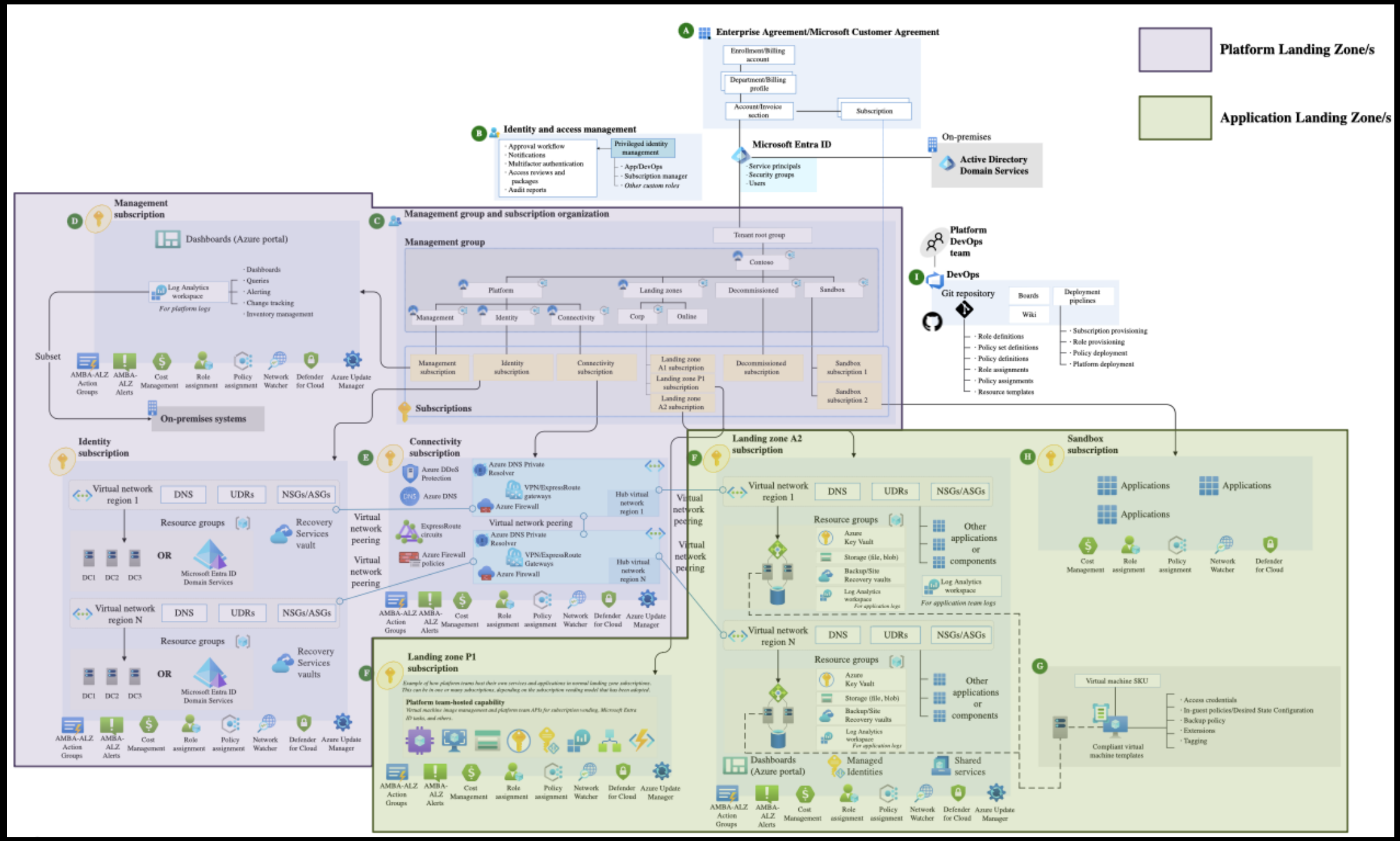

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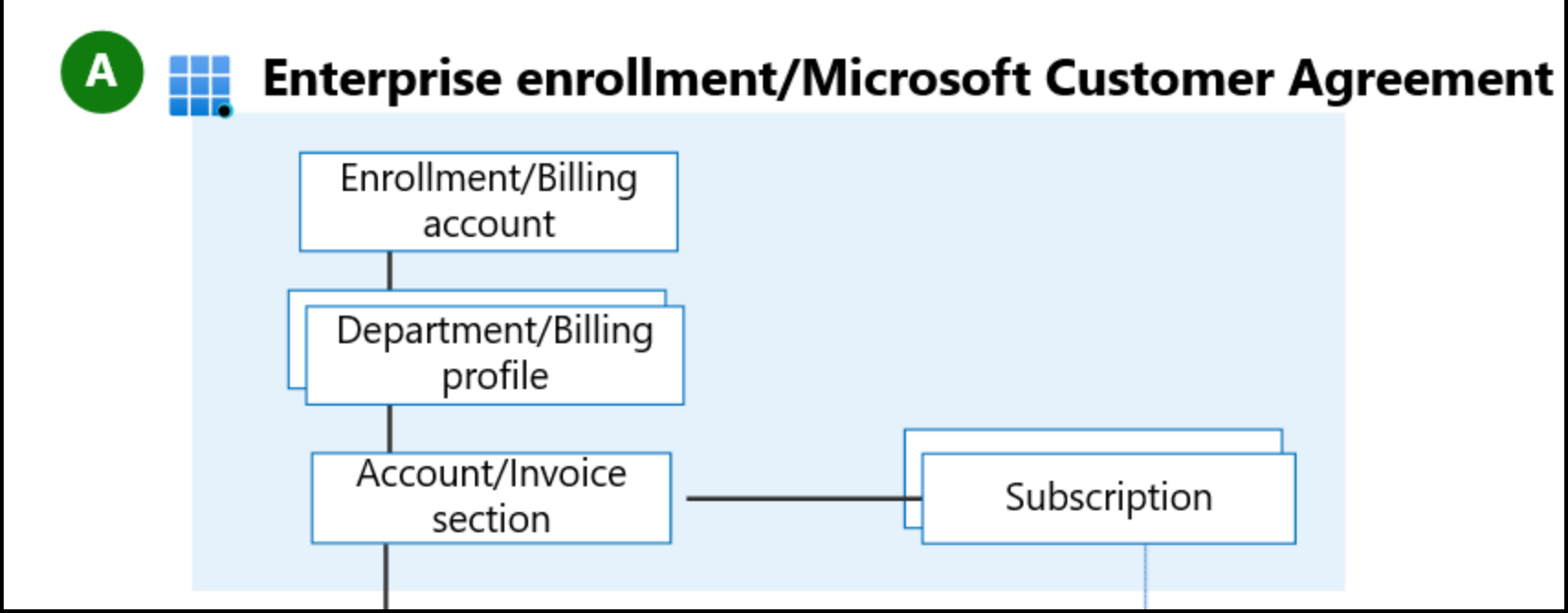
Enterprise Enrollment



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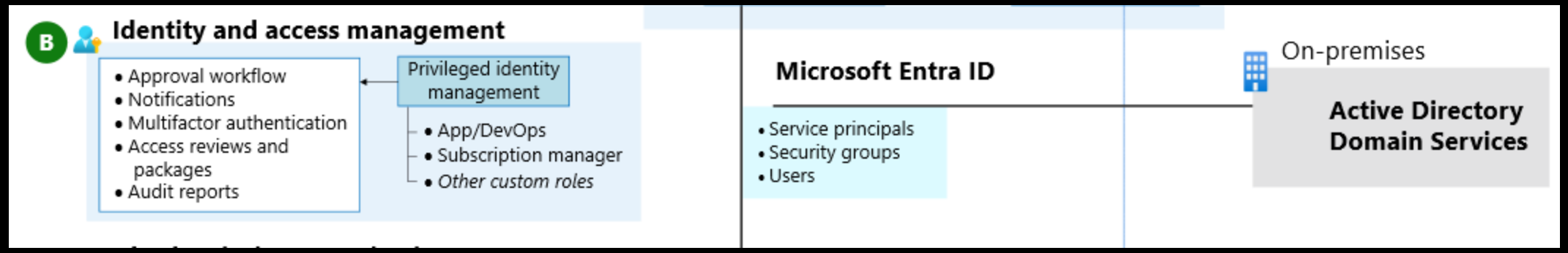
Identity and Access Management



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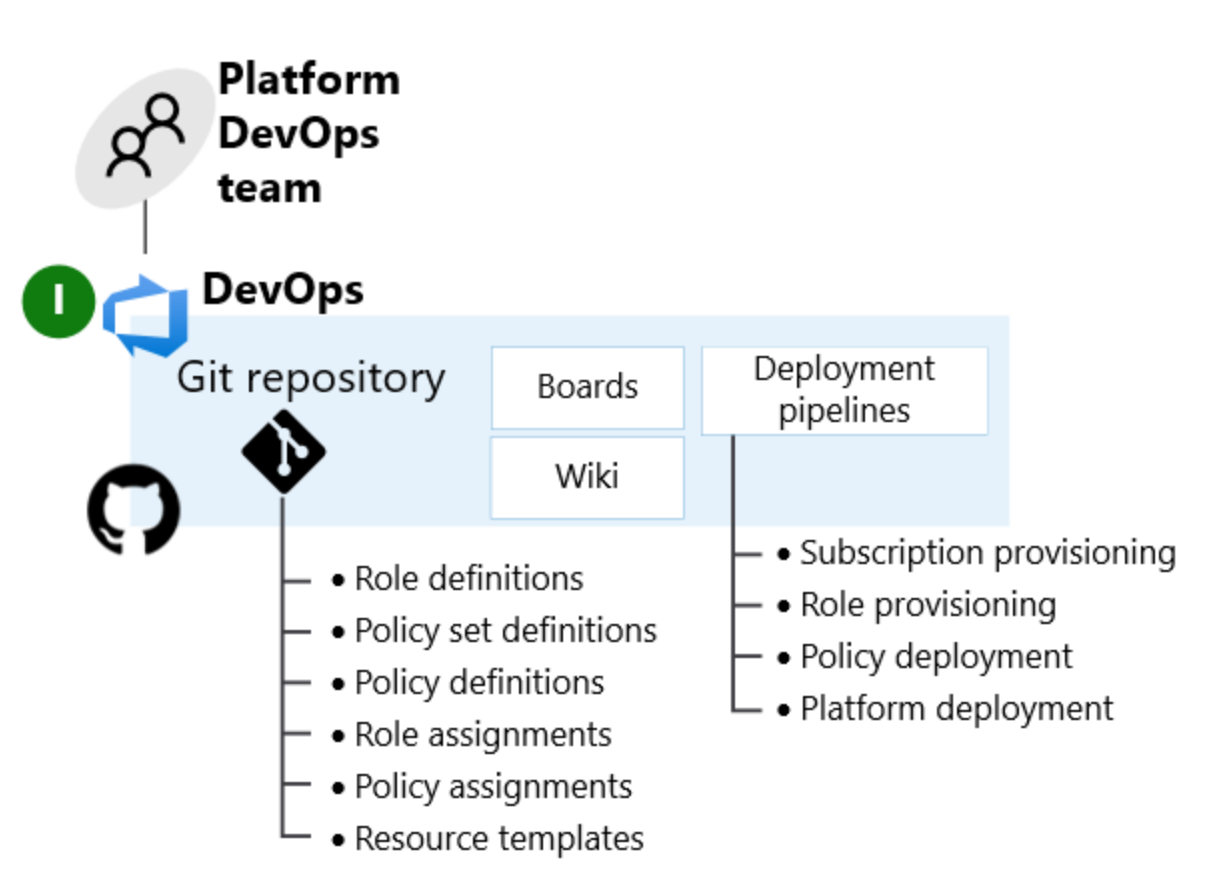
Platform automation & DevOps



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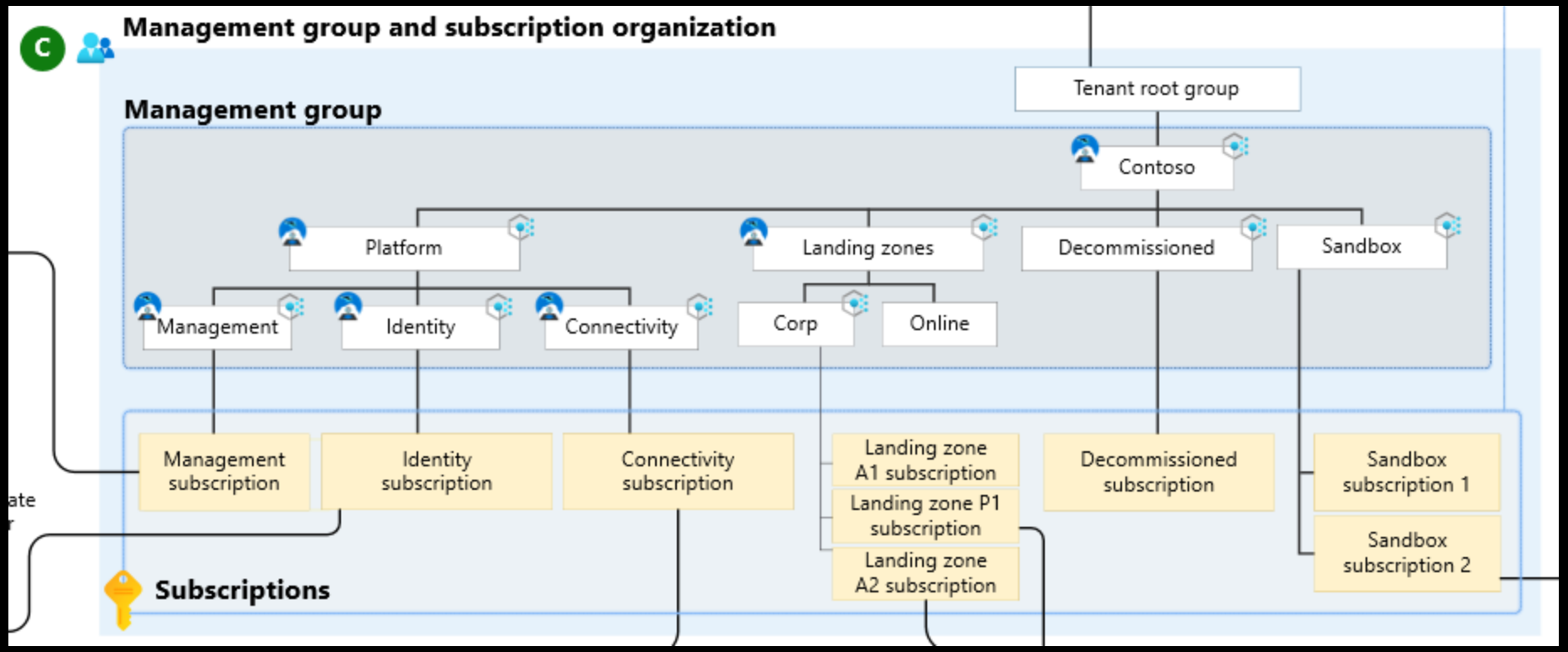
Management Group and Subscription Organization



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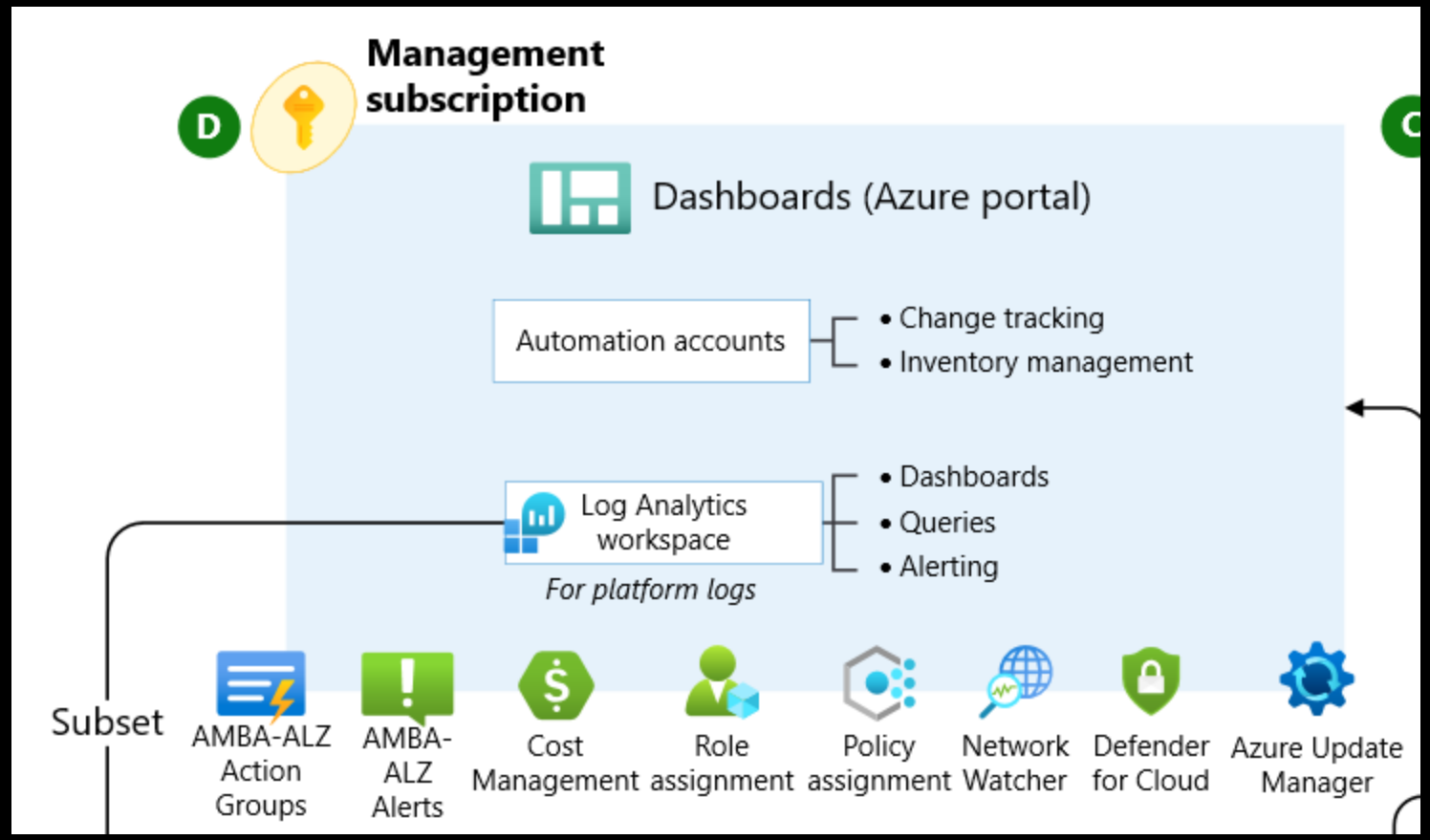
Management Subscription



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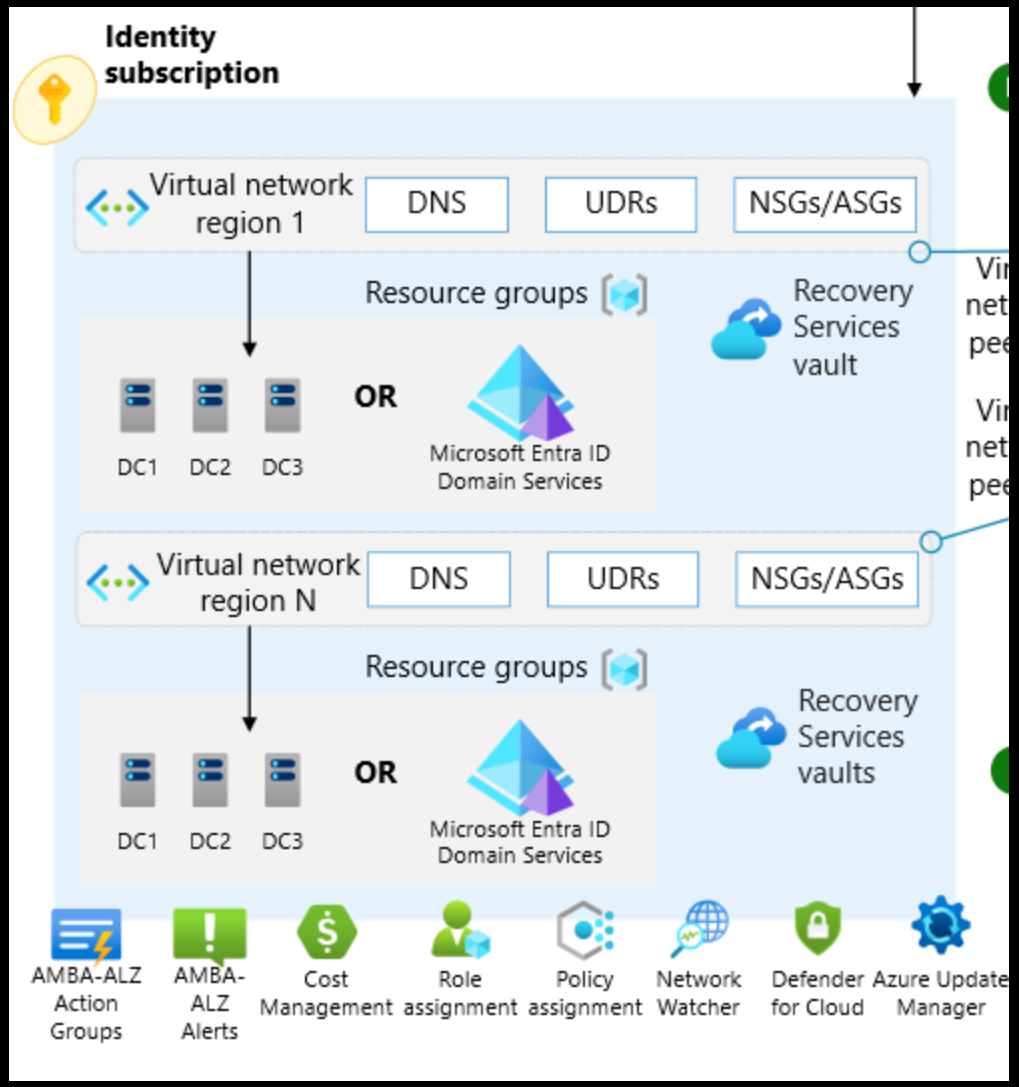
Identity Subscription



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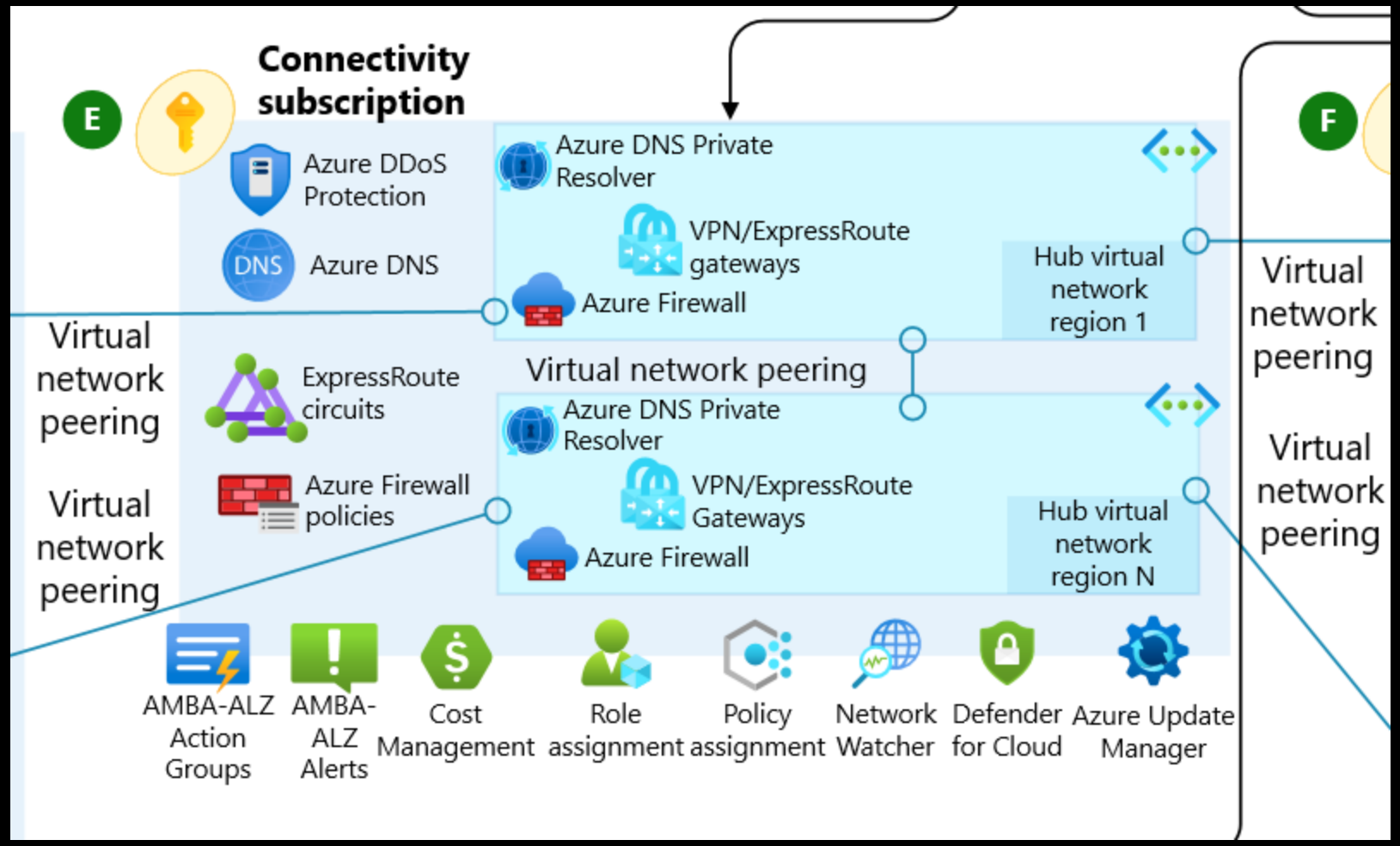
Connectivity Subscription



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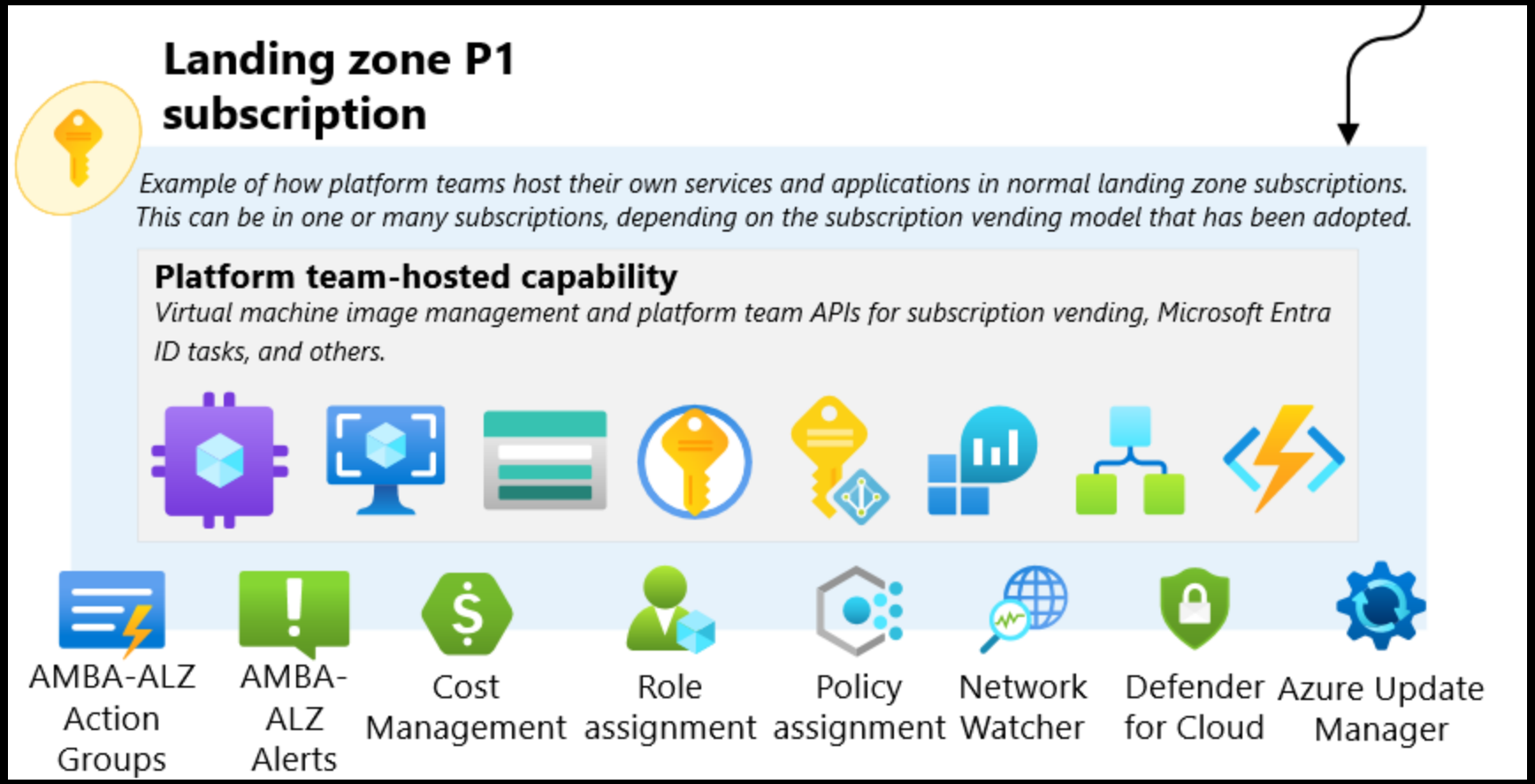
Landing Zone – P1



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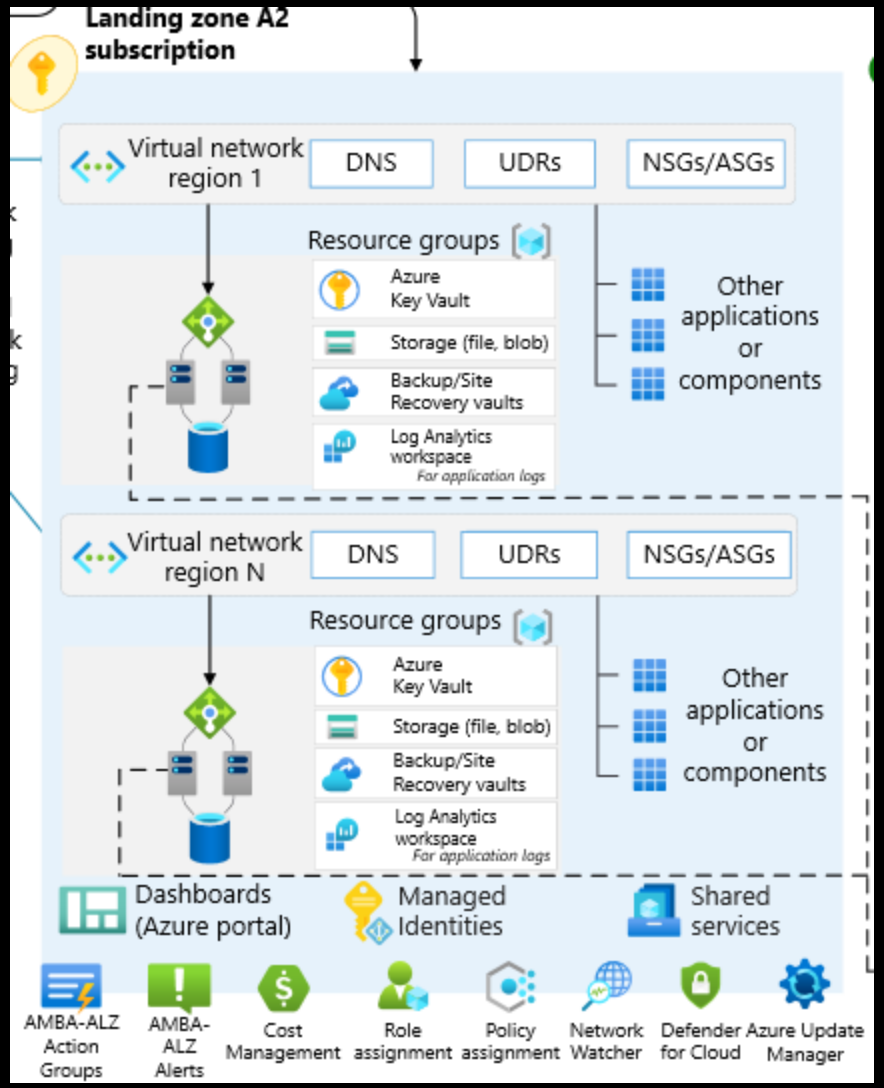
Landing Zone – A2



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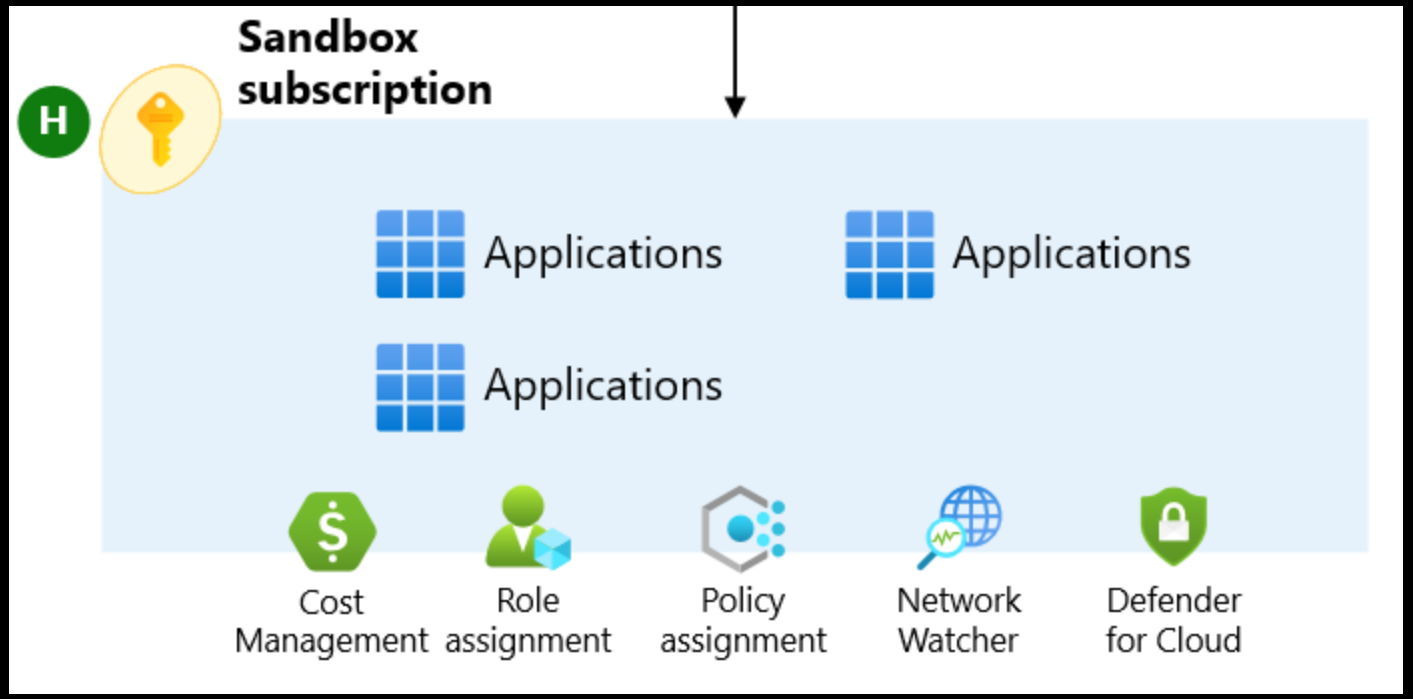
Sandbox Subscription



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